

Homework #1

Due: 11:59pm, January 23

Objectives:

- Review background from calculus and linear algebra
- Review system properties: linearity, time-invariance, causality, stability

1. **[Complex conjugate properties]**

For complex number $z \in \mathbb{C}$, let z^* denote its complex conjugate. That is, if $z = x + jy$ for $x, y \in \mathbb{R}$, then $z^* = x - jy$. Prove the following identity:

$$(\exp z)^* = \exp(z^*).$$

2. **[Eigenvalue/eigenvector]**

We consider a pair of a matrix and a column vector given by

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} e^{j0} \\ e^{j2\pi/5} \\ e^{j4\pi/5} \\ e^{j6\pi/5} \\ e^{j8\pi/5} \end{bmatrix}.$$

Show that $\mathbf{x}$ is an eigenvector of $\mathbf{A}$, that is, find a scalar λ such that $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$.

3. **[Upsampling]**

Consider the discrete-time system given by

$$y[n] = \begin{cases} x[n/2] & \text{if } n/2 \in \mathbb{Z} \\ 0 & \text{otherwise.} \end{cases}$$

Determine whether the system satisfies each of the following properties: linearity, time-invariance, causality, and stability. Justify your answer.

4. **[Convolutional systems]**

A continuous-time *convolutional* system is defined with a fixed signal $h(t)$ as

$$H\{x(t)\} = \int_{-\infty}^{\infty} x(\lambda)h(t-\lambda)d\lambda.$$

- What is the output when your input is the unit impulse, i.e., $x(t) = \delta(t)$?
- Prove the linearity of the system by showing that the identity below holds:

$$H\{ax_1(t) + bx_2(t)\} = aH\{x_1(t)\} + bH\{x_2(t)\}, \quad \forall a, b \in \mathbb{R}, \forall x_1(t), x_2(t).$$

(c) Prove that the system is time-invariant by showing that the identity below holds:

$$H\{\delta(t-d) * x(t)\} = \delta(t-d) * H\{x(t)\}, \quad \forall d \in \mathbb{R}, \forall x(t).$$

(Note that $\delta(t-d) * x(t) = x(t-d)$.)

5. [Elementary operators (Prandoni-Vetterli Exercise 3.1)]

An *operator* $\mathcal{H}$ is a transformation of a given signal and is indicated by the notation

$$y[n] = \mathcal{H}\{x[n]\}.$$

For instance, the delay operator $\mathcal{D}$ is indicated as

$$\mathcal{D}\{x[n]\} = x[n-1]$$

and the one-step difference operator is indicated as

$$\mathcal{V}\{x[n]\} = x[n] - \mathcal{D}\{x[n]\} = x[n] - x[n-1]. \quad (1)$$

A *linear* operator is one for which the following holds:

$$\mathcal{H}\{a[x[n]]\} = a\mathcal{H}\{x[n]\} \quad \mathcal{H}\{x[n] + y[n]\} = \mathcal{H}\{x[n]\} + \mathcal{H}\{y[n]\}.$$

- (a) Show that the delay operator $\mathcal{D}$ is linear.
- (b) Show that the differentiation operator $\mathcal{V}$ is linear.
- (c) Show that the squaring operator $\mathcal{S}\{x[n]\} = x^2[n]$ is not linear.

In $\mathbb{C}^N$, any linear operator on a vector $\mathbf{x}$ can be expressed as a matrix-vector multiplication for a suitable $N \times N$ matrix $\mathbf{A}$. In $\mathbb{C}^N$, we define the delay operator as the left circular shift of a vector:

$$\mathcal{D}\{\mathbf{x}\} = \begin{bmatrix} x_{N-1} \\ x_0 \\ x_1 \\ \vdots \\ x_{N-2} \end{bmatrix} \quad \text{where} \quad \mathbf{x} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{N-2} \\ x_{N-1} \end{bmatrix}.$$

Assume $N = 4$ for convenience; it is easy to see that

$$\mathcal{D}\{\mathbf{x}\} = \underbrace{\begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{\mathbf{D}} \mathbf{x}$$

so that $\mathbf{D}$ is the matrix associated to the delay operator. (Note that $\mathcal{D}$, in calligraphic style, denotes a linear operator, whereas $\mathbf{D}$, in boldface, denotes a matrix.)

- (d) Using the same definition for the one-step difference operator as in (1), write out the associated matrix for the operator in $\mathbb{C}^4$.
- (e) Consider the following matrix:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Which operator do you think it is associated to? What does the operator do?